

Pierre-Francois Roux

PhD in Genetics and Genomics | Qualified in CNU sections 64 and 65

IRCM - Inserm U1194, Molecular Oncogenesis, Montpellier, France

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RESEARCH PROFILE

I study how metabolic states shape gene regulation, chromatin organization and cell fate decisions in aging, cancer and skin biology. My work combines experimental molecular biology with functional genomics, multi-omics integration, bioinformatics, systems biology and machine-learning approaches. My current research focuses on p53/E4F1-mediated metabolic-epigenetic coupling during senescence and on ATAC-seq-derived signatures of aging, frailty and tumor evolution.

PROFESSIONAL EXPERIENCE

2021-present Research Scientist, Molecular Oncogenesis, IRCM - Inserm U1194, Montpellier.

p53-mediated metabolic reprogramming during senescence and aging; E4F1, pyruvate metabolism, acetyl-CoA, chromatin acetylation, ATAC-seq, CUT&RUN, RNA-seq, metabolomics and in vivo aging models.

2019-2021 Research Scientist, Upstream Skin Research & Innovation, Johnson & Johnson, Val-de-Reuil.

Integrative microbiome/metabolome studies of skin physiology, skin maturation, barrier function, inflammatory risk and translational biomarker discovery.

2015-2019 Postdoctoral Researcher, Institut Pasteur / Inserm U993, Nuclear Organization and Oncogenesis, Paris.

Time-resolved multi-omic profiling of cellular senescence; enhancer dynamics, AP-1 regulatory circuits, chromatin remodeling, transcriptomics, metabolomics and network inference.

2011-2014 PhD, INRAe - Institut Agro 1348 PEGASE, Rennes.

Quantitative genetics, whole-genome sequencing, QTL fine mapping, cis-eQTL analysis and RNA editing in complex metabolic traits.

EDUCATION AND TRAINING

- 2022 Animal experimentation training, designer level, BioCampus Montpellier.
- 2014 PhD in Genetics and Genomics, Institut Agro Rennes-Angers.
- 2011 MEng in Agronomical Science, Institut Agro Rennes-Angers.
- 2011 MSc in Cell and Molecular Biology, Universite Rennes 1.
- 2009 MSc in Environment and Oceanology, Universite Lille 1.
- 2008 BSc in Biology of Organisms and Populations, Universite Lille.

RESEARCH CONTRIBUTIONS

- p53/E4F1, metabolism and senescence: investigated how E4F1 connects p53/pRB signaling, pyruvate metabolism, acetyl-CoA production and chromatin regulation during senescence.
- ATAC-seq as an integrative assay: established bulk ATAC-seq as a dual genetic and epigenetic platform capturing chromatin accessibility, variants, CNVs, mitochondrial features and telomere-associated signals.
- Systems view of senescence: developed temporal multi-omic analyses identifying enhancer remodeling and AP-1-centered regulatory circuits as key determinants of oncogene-induced senescence.
- Skin microbiome and metabolome: integrated microbiome and metabolome profiles to identify microbe-metabolite clusters relevant to skin physiology and early-life skin maturation.
- Quantitative genetics: combined QTL mapping, WGS, selection signatures, variant annotation and eQTL analyses to prioritize causal genes for adiposity and energy metabolism.
- RNA editing: developed stringent WGS/RNA-seq strategies to distinguish genuine RNA editing events from sequencing and mapping artifacts.

PUBLICATIONS AND VALORIZATION

- 24 original research articles, including 7 first or co-first author publications and 1 last/corresponding author publication.
- 2 review articles, 1 didactic article, 1 radio communication and 4 patents.
- Major journals include NAR Genomics and Bioinformatics, Nature Communications, Nature Aging, Journal of Investigative Dermatology, Nature Cell Biology, Nature, Cell and Cell Genomics.

SELECTED PUBLICATIONS

- Toumi I., Kham C., Stuani L., Le Cam L., Roux P.F. Beyond chromatin accessibility: bulk ATAC-seq as an integrative assay to portray genomes and epigenomes. *NAR Genomics and Bioinformatics*, 2026. Last and corresponding author.
- Di Michele M., Attina A., Roux P.F. et al. E4F1 coordinates pyruvate metabolism and the activity of the elongator complex to ensure translation fidelity during brain development. *Nature Communications*, 2025. Co-second author.
- Guerrero A. et al. 3-Deazaadenosine alleviates senescence to promote cellular fitness and cell therapy efficiency in mice. *Nature Aging*, 2022.
- Roux P.F., Oddos T., Stamatias G. Deciphering the role of skin surface microbiome on skin health: an integrative multi-omics approach reveals three microbial/metabolic clusters. *Journal of Investigative Dermatology*, 2021. First author.
- Martinez-Zamudio R.I., Roux P.F. et al. AP-1 imprints a reversible transcriptional programme of senescent cells. *Nature Cell Biology*, 2020. Co-first author.
- Heinzmann F. et al. Necroptosis microenvironment determines lineage commitment in liver cancer. *Nature*, 2018.
- Martinez-Zamudio R.I., Robinson L., Roux P.F., Bischof O. Snapshot: cellular senescence pathways / in *pathophysiology*. *Cell*, 2017.
- Roux P.F. et al. Combined QTL and selective sweep mappings with coding SNP annotation and cis-eQTL analysis revealed PARK2 and JAG2 as candidate genes for adiposity regulation. *G3*, 2015. First author.

FUNDING

- Submitted: ANR 2026 project on metabolism-epigenome interactions during cellular senescence and aging.
- Submitted: Cancerpole Grand Sud-Ouest 2026 project on integrative tumor profiling by ATAC-seq for actionable signatures.
- Acquired: ANR 2019 S-ENCODE project on deciphering the senescence code to improve healthy lifespan.

TEACHING

- Universite de Montpellier: Master 2 Biology-Health, molecular basis of cellular senescence and aging, lecture, 4 h, 2021.
- Institut Agro Rennes-Angers: engineering students and continuing education, RNA-seq, ChIP-seq, ATAC-seq, UNIX/R and genomic data analysis, since 2014.
- Universite Rennes 1: Master 1 Cell and Molecular Biology, practical/statistical training, 16 h, 2012.
- SIRIC Montpellier Cancer: Unlocking epigenomic insights: ATAC-seq data analysis in practice, practical training, since 2025.

SUPERVISION AND TRAINING

- Co-supervision of 1 PhD student on aging, cellular senescence and metabolic adaptability.
- Supervision of 7 Master 2 internships in bioinformatics, epigenomics, cancer biology and structural genomics.
- Supervision of 2 Master 1 internships in molecular and cellular biology.
- Member of a PhD thesis committee.

SCIENTIFIC ANIMATION, RESPONSIBILITIES AND SERVICE

- Invited seminars at international research institutes on systems views of senescence and metabolic reprogramming.
- Participation in conference organization, scientific events and collaborative research networks.
- Editorial and peer-review activities for scientific journals.
- Involvement in professional equality and mentoring actions, including Femmes & Sciences mentoring.

TECHNICAL SKILLS

Functional genomics; RNA-seq; ATAC-seq; CUT&RUN; ChIP-seq; MNase-seq; WGS; QTL mapping; cis-eQTL; metabolomics; microbiome profiling; UNIX; R/Bioconductor; reproducible workflows; multi-omics integration; network inference; dimensionality reduction; feature selection; machine learning; cellular senescence models; primary cells; mouse models.